

Ex.No.10 – Use Ghidra to Disassemble and Analyze an Executable

Aim

To use Ghidra to disassemble and analyze an executable file, examine its functions, strings, imported components, and decompiled code, and understand the basic behavior of the analyzed program.

Objectives

- To understand the basic use of Ghidra for executable analysis.
- To examine the structure and components of a Portable Executable (PE) file.
- To identify strings, imported components, and functions using Ghidra.
- To examine the main function using the CodeBrowser and Decompiler.
- To understand the basic workflow used in static malware analysis.

Tools Required

- Ghidra 12.1.4
- Java Development Kit (JDK)
- Windows operating system
- Benign test executable (sample.exe)

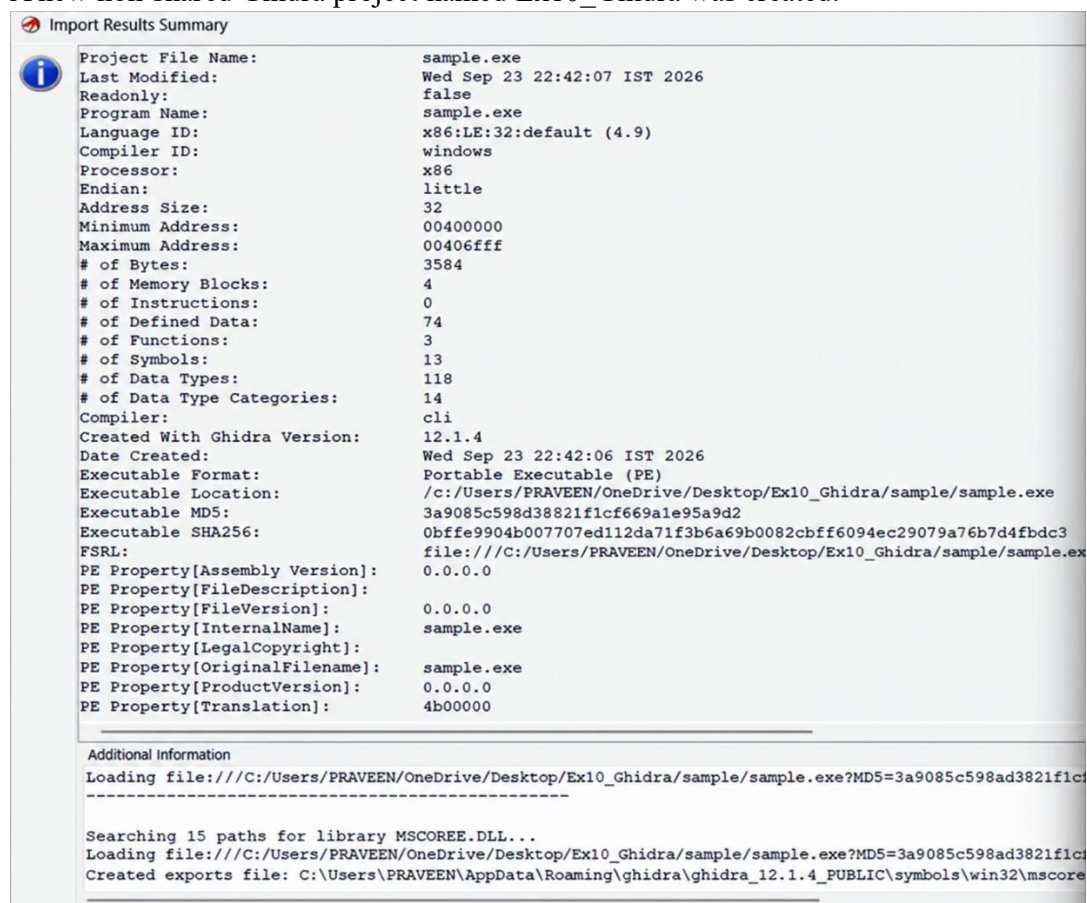
Safety Note

A harmless test executable was used for this experiment instead of live malware. The executable was analyzed statically without intentionally executing it.

Procedure

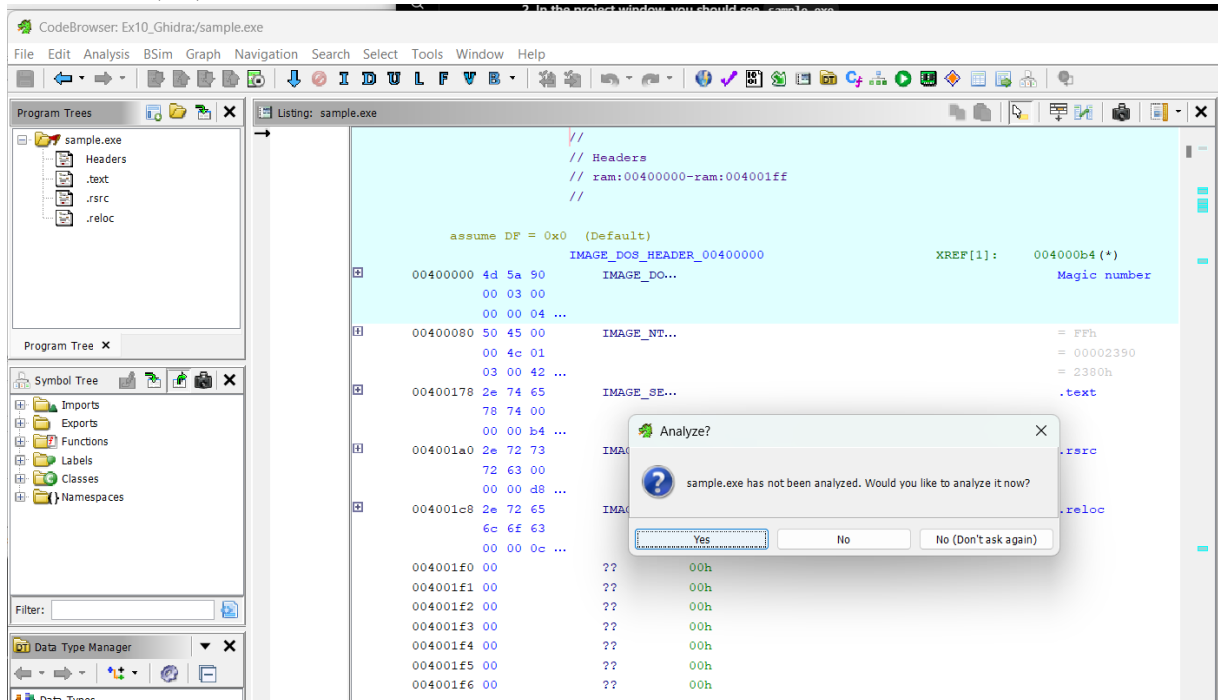
Step 1 – Create a Ghidra Project

A new non-shared Ghidra project named Ex10_Ghidra was created.



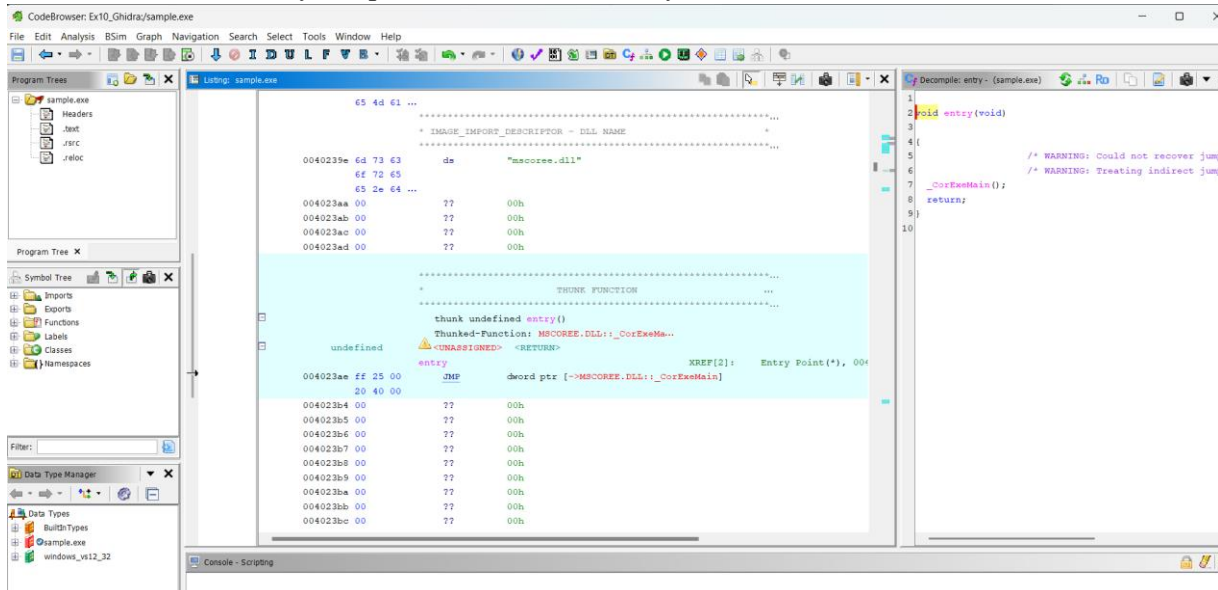
Step 2 – Import the Executable

The sample.exe file was imported into the Ghidra project. Ghidra identified the file as a Portable Executable (PE).



Step 3 – Perform Automatic Analysis

The default Ghidra analysis options were used to analyze the executable.



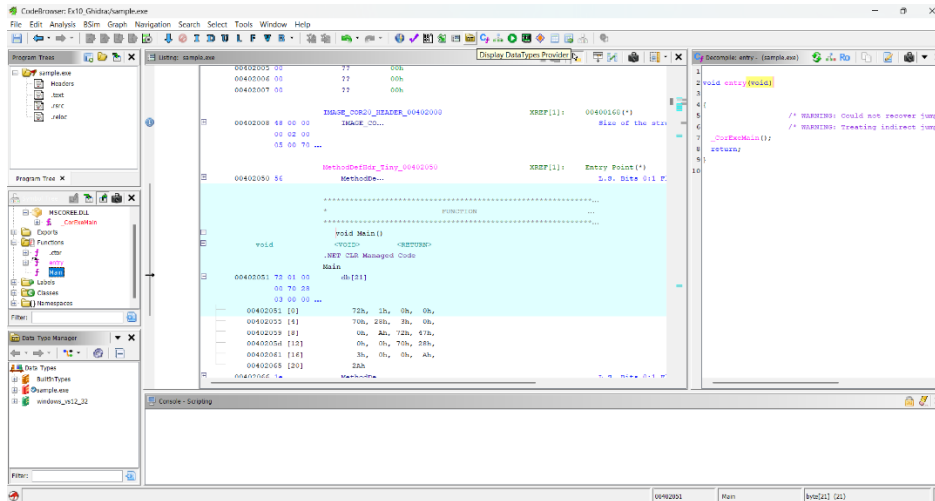
Step 4 – Analyze Strings

The String Search feature was used to identify readable strings in the executable. Ghidra identified 44 strings.

Important strings included:

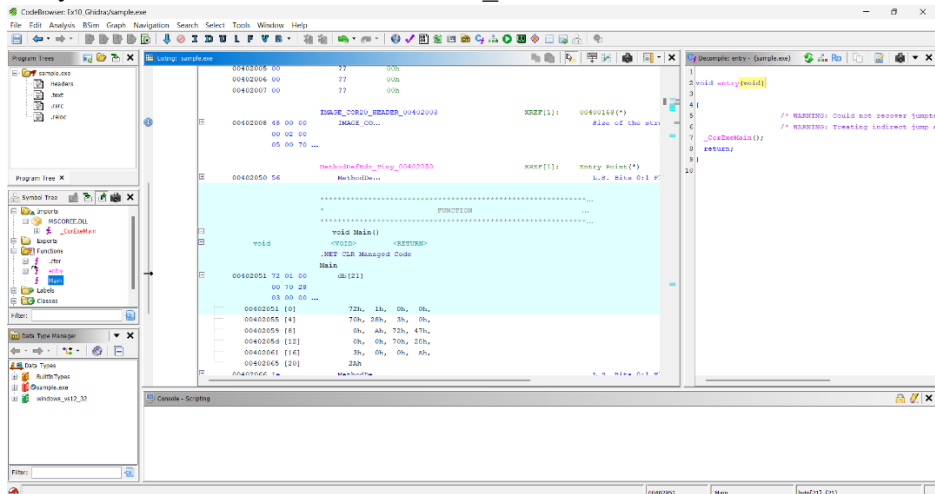
- Ghidra Malware Analysis Lab Sample
- This is a harmless test program.
- Console

- WriteLine



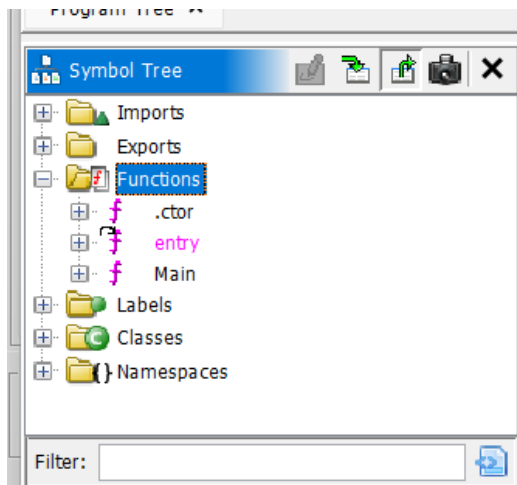
Step 5 – Analyze Imported Components

The Imports section was examined to identify external components used by the executable. The analysis showed MSCORE...DLL and _CorExeMain.



Step 6 – Analyze Functions

The Functions section was examined. Ghidra identified the functions .ctor, entry, and Main.



Step 7 – Analyze the Main Function

The Main function was opened in the Ghidra CodeBrowser and examined using the available decompiler view.

Observation

The executable was successfully loaded and analyzed using Ghidra. The file was identified as a Portable Executable (PE) for the x86 architecture. Ghidra identified the .text, .rsrc, and .reloc sections and detected 44 readable strings. The imported runtime component included MSCORE...DLL with _CorExeMain. The Functions window identified .ctor, entry, and Main.

The strings found in the executable indicated that it was a harmless test program created for analysis purposes.

Rubrics

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result

The sample.exe file was successfully imported, disassembled, and analyzed using Ghidra. The executable's strings, imported components, functions, and Main function were examined using the Ghidra CodeBrowser and available decompiler views.